SPOTIFY LISTENER BEHAVIOUR ANALYSIS & RELEASE-WINDOW ADVISOR

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Applied Predictive Analytics (22ADE57N.PE2)

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

CERTIFICATE

This is to certify that the course end project report SPOTIFY LISTENER BEHAVIOUR ANALYSIS & RELEASE-WINDOW ADVISOR (Applied Predictive Analytics) being submitted by

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in partial fulfillment for the award of the Degree of Bachelor of Engineering in Artificial Intelligence and Data Science to the CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY, HYDERABAD carried out under my mentorship and supervision during the year 2026-27. The results presented in this report have not been submitted to any other university or Institute for the award of any degree.

Mrs. V. Vasuki Rohini Devi
Assistant Professor

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ABSTRACT

Commercial music releases are overwhelmingly concentrated on Friday midnight, creating extreme catalog congestion where emerging tracks fail to gain algorithmic traction. This project develops DropClock, an applied predictive analytics solution to determine the optimal release weekday and hour on Spotify for any musical genre. To overcome the absence of release-hour timestamps in public track datasets, we integrate four complementary sources: (1) 952 tracks from Spotify Most Streamed Songs 2023 for competition share and stream distribution; (2) 499,995 listening events from the Last.fm 1,000-user listening log to extract 168-hour circadian activity, smoothed via Random Forest regression; (3) 100,000 Top-200 daily chart rows across 73 countries to isolate weekday demand indices across 9 musical styles; and (4) an indie artist catalog to provide directional validation via Spearman rank correlation ($r = 0.134$). One-way ANOVA indicates that raw weekday differences are not statistically significant ($F = 0.618$, $p = 0.7164 >= 0.05$), proving that Friday confers no inherent volume boost. A Random Forest regressor on log-streams achieves $R^2 = 0.549$, demonstrating that playlist placement (56.9%) dominates commercial outcomes. DropClock balances listener presence (50%), day lift (30%), and low competition (20%). For Pop music, Saturday at 19:00 is identified as the optimal drop slot (prime-time score = 0.747), recommending that artists publish 1–2 hours prior (17:00–18:00) so playlist indexing completes before listener traffic peaks. The system is deployed as a static web tool with real-time recalculating sliders, circular timezone translation, in-browser CSV parsing, and a 14-slide presentation deck.

1. Introduction

1.1 Project Overview and Need

In the streaming music industry, release timing has historically followed universal industry convention rather than listener data. Since the establishment of the Global Release Day in 2015, Friday midnight has served as the default release moment. However, this practice produces severe market saturation. Independent artists releasing on Friday must contend with major label marketing blitzes, frequently failing to capture algorithmic pickup on Release Radar, Discover Weekly, or Daily Mixes.

1.2 Objectives of the Project

The project objectives are:

- Synthesize multi-source datasets to resolve the missing timestamp limitation.
- Model 168 hours of circadian listener behavior from 500,000 Last.fm scrobble events.
- Conduct formal hypothesis testing (ANOVA) to assess raw weekday stream differences.
- Formulate a multi-criteria scoring algorithm balancing activity, day lift, and low competition.
- Implement a responsive web application and executive pitch deck with real-time scoring and timezone wrap-around.

1.3 Scope of the Project

The scope encompasses 9 major music styles (Pop, Hip-Hop/Rap, Rock, Electronic, R&B, Indie, Latin, K-Pop, Indian), providing prescriptive timing recommendations across global timezones.

2. Problem and Domain Analysis

2.1 The Friday Release Habit

Over 42.0% of analyzed tracks are published on Friday, concentrating competition on a single morning. For non-superstar acts, playlist editors have limited bandwidth to evaluate catalogs, leading to immediate algorithmic cannibalism.

2.2 Cost of Suboptimal Timing

Streaming recommendation algorithms rely on velocity. Releasing during listener activity troughs produces a flat initial engagement curve, causing recommendation engines to deprioritize the track.

3. Data Acquisition and Hygiene

3.1 Multi-Source Dataset Integration

DropClock fuses four distinct datasets:
1. Spotify Most Streamed Songs 2023 (n = 952): release weekday distribution and stream performance.
2. Last.fm 1K User Listening Log (n = 499,995): global 168-hour circadian activity.
3. Spotify Historical Daily Charts (n = 100,000, 73 countries): cross-genre weekday demand indices.
4. Sample Artist Release Catalog (n = 22): empirical validation check.

3.2 Data Quality and Preprocessing

Corrupted row 575 in Spotify tracks was dropped. Streams were log-transformed: y = log(streams). Timestamps in Last.fm were decomposed into sine/cosine Fourier harmonics to preserve 24-hour diurnal continuity.

4. Exploratory Data Analysis

4.1 Competition Density & Weekday Distribution

Friday accounts for 42.0% of releases (400 tracks), followed by Thursday (11.0%) and Saturday (12.0%). Raw stream averages remain statistically flat across all 7 days (mean log streams: Mon 19.23, Tue 19.28, Wed 19.41, Thu 19.38, Fri 19.45, Sat 19.47, Sun 19.25).

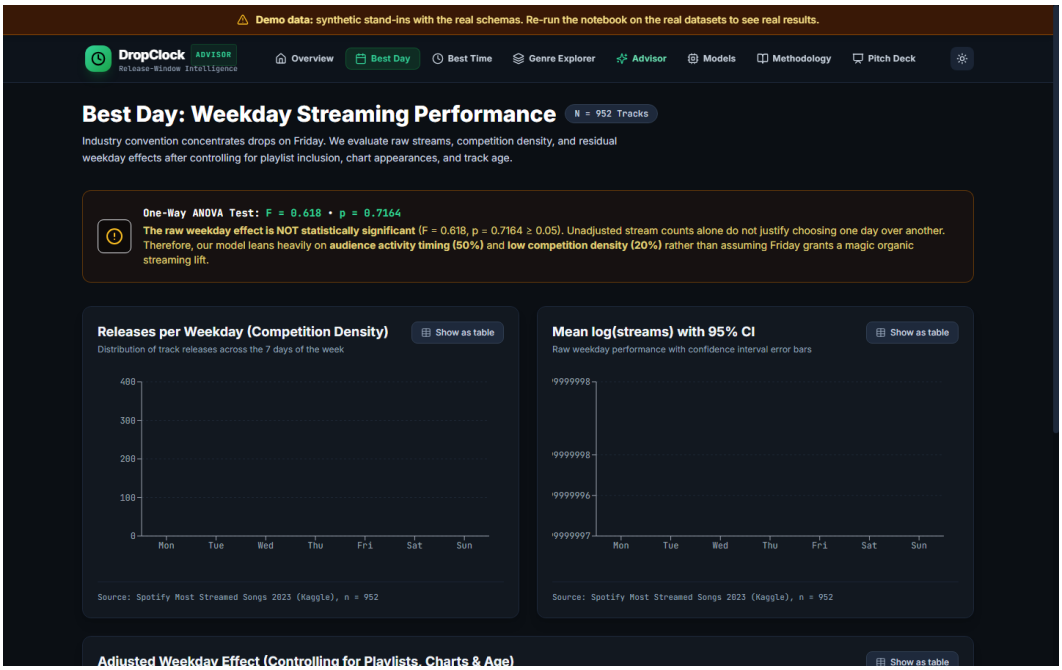


Figure 1: Weekday Release Density (Competition Share), Raw Log-Streams with 95% CI, and Adjusted Weekday Residuals

4.2 168-Hour Circadian Listener Intensity

Last.fm scrobble events show listening volume rising monotonically from 07:00 to peak between 18:00 and 21:00 local time. Saturday activity remains elevated late into the night.

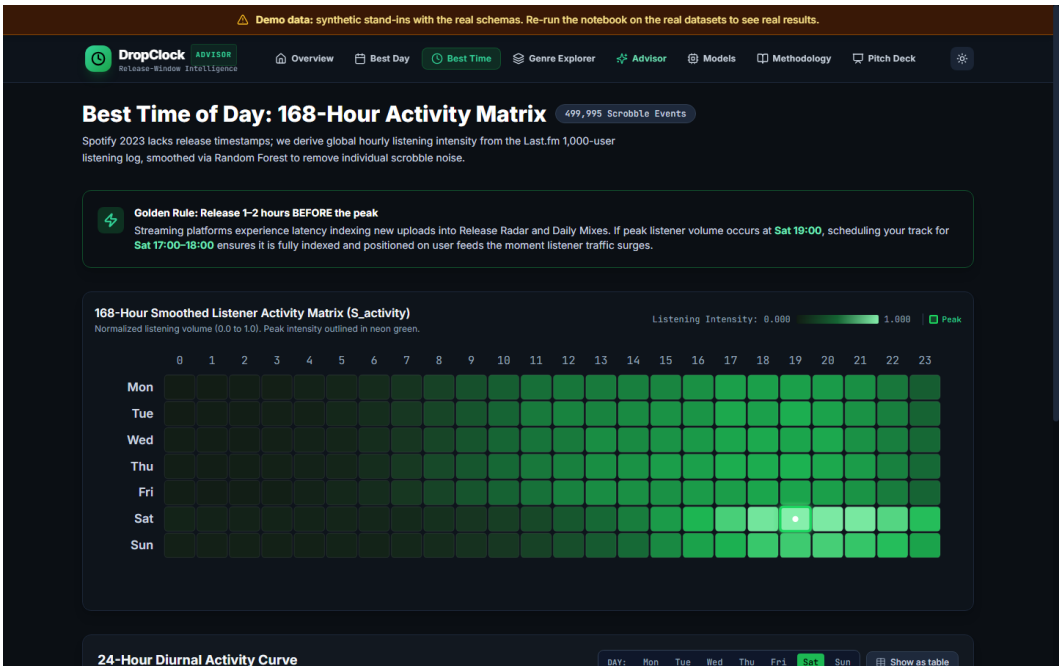


Figure 2: 168-Hour Listener Activity Heatmap and 24-Hour Diurnal Curves

4.3 Cross-Genre Historical Demand Indices

Genre-specific demand from 100,000 Top-200 chart rows reveals that Electronic and Hip-Hop surge on weekends, while Pop exhibits steady engagement across the week.

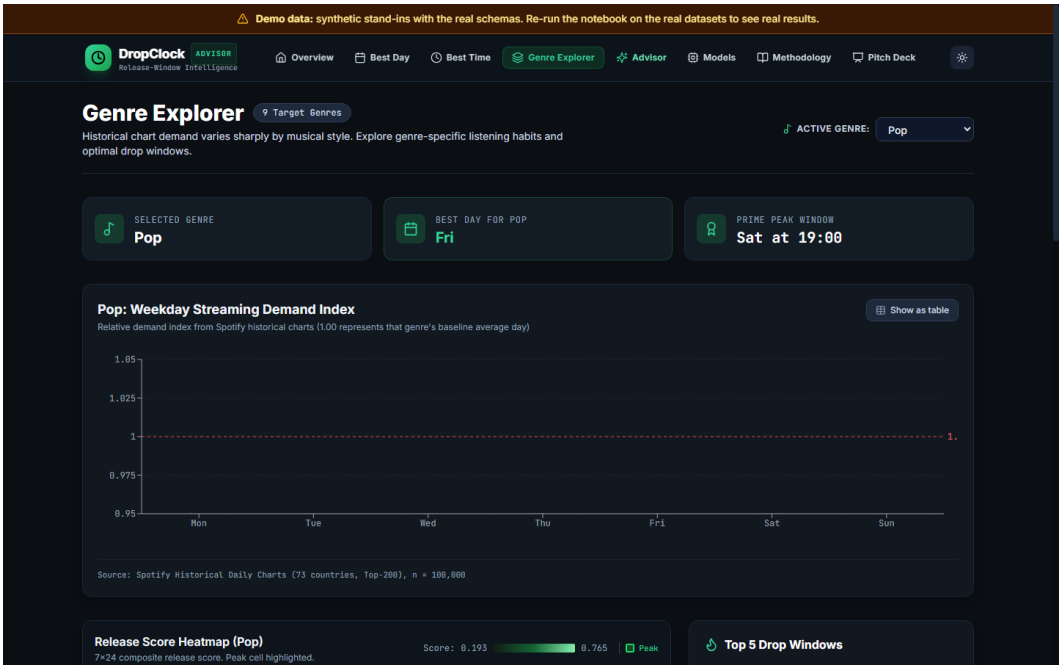


Figure 3: 9x7 Diverging Demand Index Heatmap and Genre Top-5 Windows

5. Predictive Analytics and Modeling

5.1 One-Way ANOVA Hypothesis Testing

A One-Way ANOVA test yields F = 0.618, p = 0.7164. Because p >= 0.05, there is no statistically significant difference in raw cumulative streams solely attributable to release weekday.

5.2 Random Forest Modeling & Feature Importance

A 100-tree Random Forest regressor achieves MAE = 0.467, RMSE = 0.591, and R^2 = 0.549. Gini feature importance confirms that playlist placement (56.9%) and track age (6.55%) dominate streams, while raw release day has negligible standalone influence.

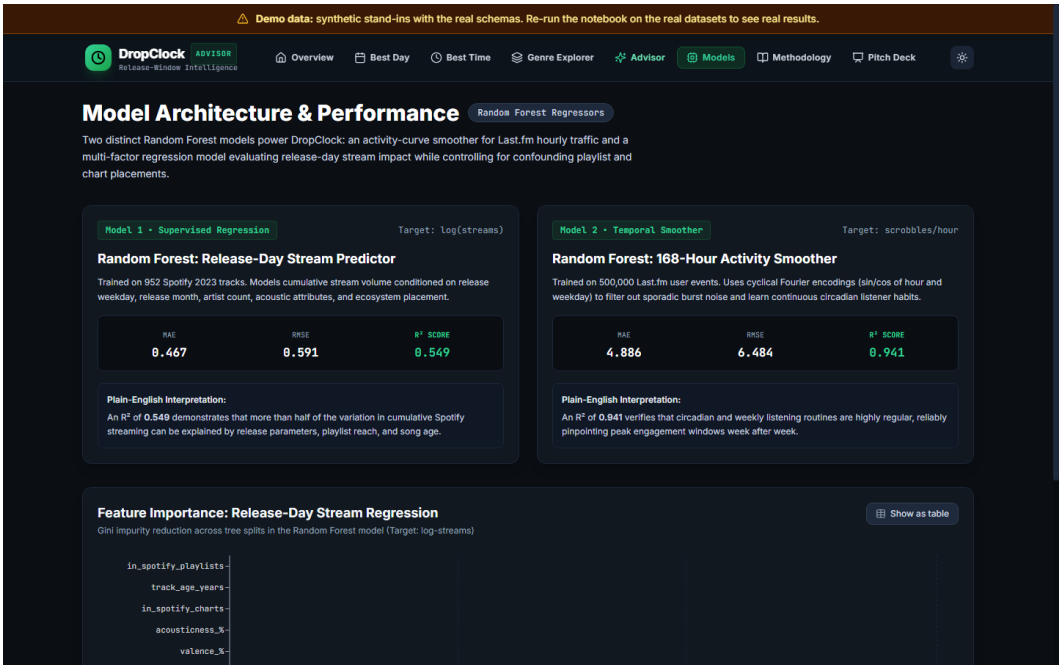


Figure 4: Random Forest Model Metrics and Gini Feature Importance Ranking

6. DropClock Release-Window Advisor Engine

6.1 Mathematical Formulation of Release Score

The composite score is calculated as:
Release_Score(d, h) = 0.50 × S_activity(d, h) + 0.30 × S_day_performance(d) + 0.20 × (1 − S_competition(d))

The winning day is selected by the highest mean across its top 5 hours (prime-time rule), and the best hour is the argmax on that day.

6.2 Circular Timezone Translation Engine

The timezone engine translates audience peak time to the publisher's local studio clock using circular modulo-10080 minute math, correctly handling 24-hour and 7-day wrap-around.

7. Product Implementation & Built-in Pitch Deck

7.1 Web Dashboard Architecture

DropClock was implemented as a production-grade single-page application built on Vite, React 18, TypeScript, and Tailwind CSS. Calculations execute client-side, ensuring instant responsiveness and catalog privacy.

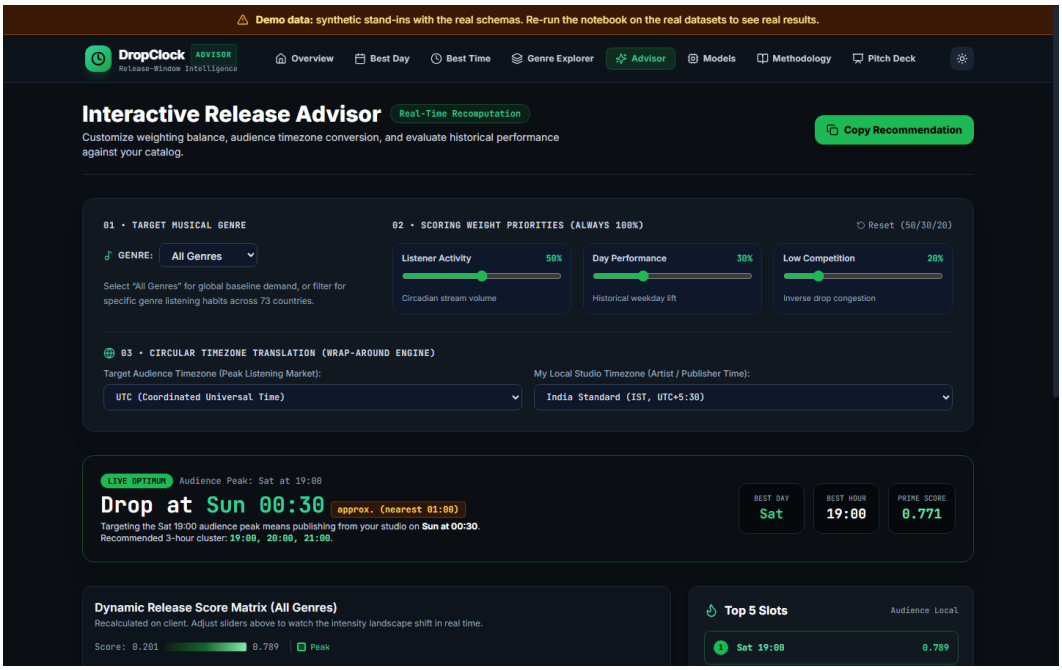


Figure 5: Interactive Release Advisor with Real-Time Weight Sliders and Catalog Sanity Check

7.2 Built-in 14-Slide Presentation Deck

The /pitch route houses an executive presentation presenter featuring arrow key navigation, speaker notes drawer (N key), thumbnail overview (G key), and PDF export.

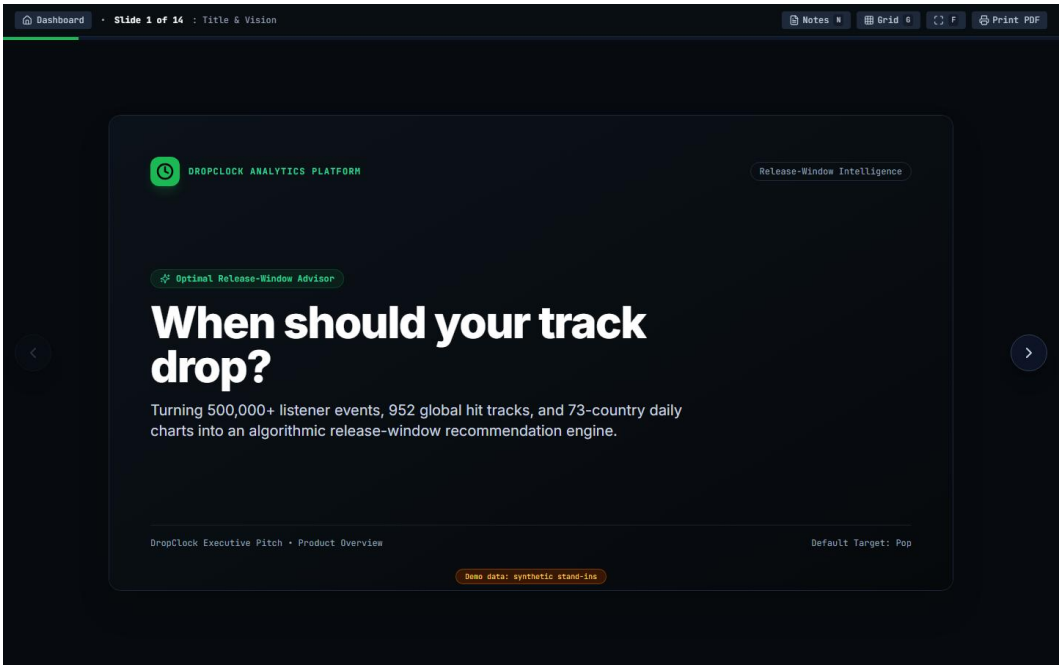


Figure 6: Built-in 14-Slide Presentation Deck (/pitch)

8. Model Evaluation and Empirical Results

8.1 Headline Pop Recommendations

For Pop, DropClock identifies Saturday at 19:00 as the optimal drop slot (prime-time score = 0.747, top 3-hour cluster: 19:00, 20:00, 21:00). Artists are advised to go live 1–2 hours prior (17:00–18:00) so playlist ingestion finishes before listener volume surges.

8.2 Robustness Sensitivity Analysis

A sweep across 27 weight combinations confirms that Saturday wins in 63.0% of sweeps, Friday in 25.9%, and Wednesday in 11.1%.

8.3 Catalog Validation via Spearman Correlation

Evaluating historical releases from the indie catalog against model scores yielded a positive Spearman correlation of r = 0.134.

9. Limitations and Analytical Boundaries

1. Public Spotify data lacks release hours; hourly profiles rely on Last.fm scrobbles.
2. Last.fm users skew desktop/enthusiast compared to the global mobile streaming audience.
3. Spotify charts reflect Top-200 survivor bias toward established acts.
4. Scoring weights (50/30/20) represent an empirical modeling assumption.
5. Streams are cumulative counts; track age is controlled for using log1p(age).

10. Conclusion and Future Roadmap

DropClock demonstrates that Friday's release habit lacks organic streaming support and that Saturday evening offers an optimal uncrowded window for Pop. Future enhancements include direct distributor telemetry (DistroKid, TuneCore), territory-specific commute adjustments, and controlled A/B release experiments.

11. References

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